

# 2 — Hardware Setup & Wiring Guide

**Audience:** Field Technician / Electrical Engineer  
**Replaces:** Root-level installation\_guide.md (now deleted)

## Parts Required Per Station

| Item                      | Specification                                | Notes                                |
|---------------------------|----------------------------------------------|--------------------------------------|
| Waveshare ESP32-S3 A7670E | 4G LTE + GPS development board               | Main controller                      |
| PZEM-004T v3.0            | AC Power Meter module                        | Sensor (100A split-core CT included) |
| 1NCE IoT SIM Card         | Micro-SIM, flat-rate €10/10 years            | Or any LTE SIM with APN credentials  |
| DIN-rail PSU              | 5V DC / 2A minimum (e.g. Mean Well MDR-10-5) | Powers the ESP32 board               |
| IP65 plastic enclosure    | Min. 150×100×70 mm with cable glands         | Houses the board inside the cabinet  |
| 4G Antenna                | External SMA whip antenna                    | Supplied with Waveshare board        |
| GNSS Antenna              | Passive or active GPS patch antenna          | Supplied with Waveshare board        |

## Section 1 — SIM Card Activation (1NCE)

1. Go to the [1NCE Customer Portal](#) and create an account.
2. Click **Activate SIM Card** → enter the **ICCID** and **Activation Code** printed on the SIM plastic holder.

3. Wait **10–30 minutes** for network allocation.
4. Insert the SIM into the **Micro-SIM slot** on the Waveshare board. Golden contacts face **downward** toward the PCB.

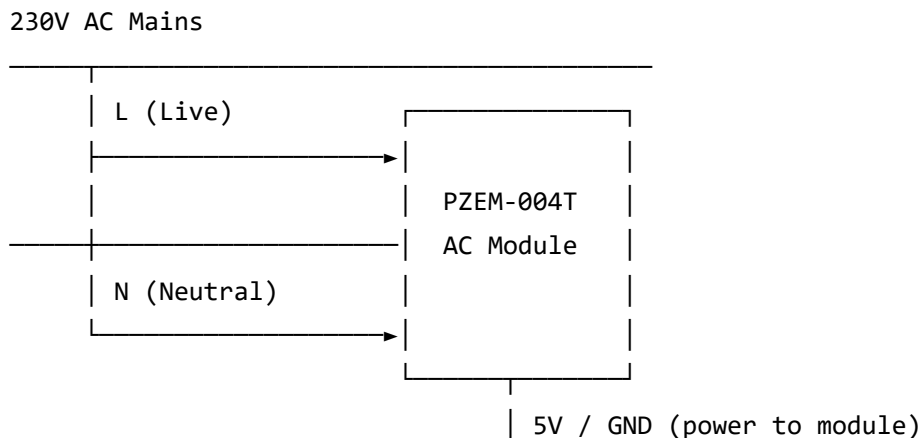
**Note:** The firmware is pre-configured for `iot.1nce.net` APN — no username or password required.

## Section 2 — PZEM-004T Wiring

The PZEM-004T communicates with the ESP32 board using **Modbus RTU over UART** (Serial2).

### 2.1 Power Connections

The PZEM-004T module needs a **5V power supply** and must be connected to the **AC line** being measured:



Connect the **PZEM-004T power input terminals** ( L and N ) to the mains supply (or to the load/pump side — either works). The module draws ~1W from the mains for self-power.

### 2.2 Current Transformer (CT Clamp)

- Clip the **included split-core CT clamp** around a **single conductor** of the pump's supply cable (L1, L2, or L3 — pick one active phase).
- Do **not** clamp around both L and N together — the magnetic fields cancel and you will read zero.
- Plug the CT into the round jack socket on the PZEM module.

## 2.3 UART Serial Data Wiring (PZEM ↔ ESP32-S3)

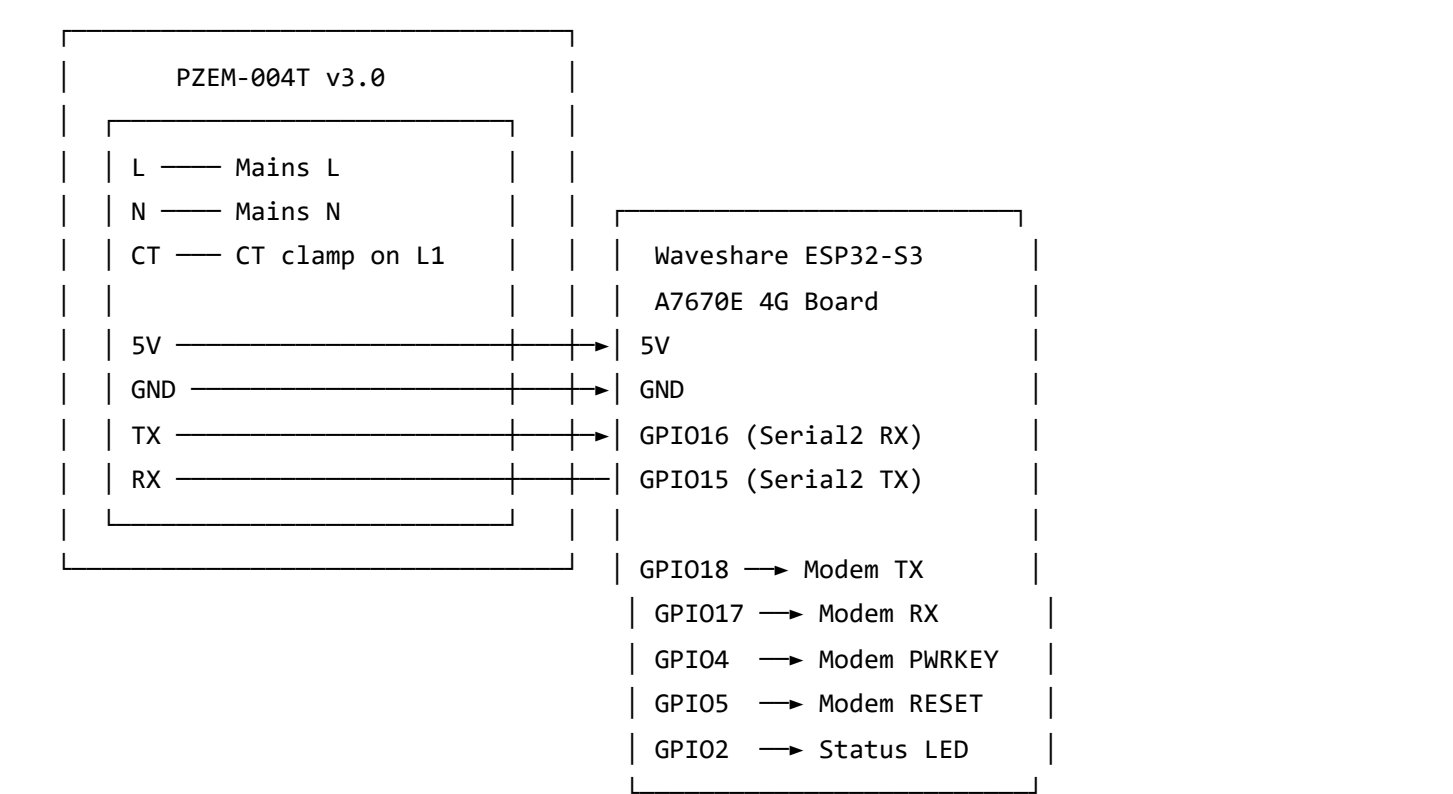
Connect the PZEM-004T TTL serial header to the ESP32-S3 board:

| PZEM-004T Pin | ESP32-S3 GPIO               | Wire Color Suggestion |
|---------------|-----------------------------|-----------------------|
| 5V            | 5V (board header)           | Red                   |
| GND           | GND (board header)          | Black                 |
| TX            | <b>GPIO 16</b> (Serial2 RX) | Yellow                |
| RX            | <b>GPIO 15</b> (Serial2 TX) | Green                 |

⚠ **Important:** The PZEM TX connects to the ESP32 RX (GPIO 16), and PZEM RX connects to ESP32 TX (GPIO 15). Cross-connect TX→RX.

⚠ **Voltage:** PZEM TTL serial operates at **3.3V logic levels** — safe for direct connection to the ESP32-S3 (also 3.3V). Do not connect to 5V-only microcontrollers without a level shifter.

## 2.4 Wiring Diagram (ASCII)



## Section 3 — Antenna Placement

### 4G Cellular Antenna

- Screw the external **SMA whip antenna** onto the gold SMA connector.
- If the main electrical cabinet is **metal**, route the antenna **outside** through a rubber-gasketed hole or place it near a non-metallic vent. A metal cabinet will block LTE signals.
- In weak signal areas, use a high-gain external whip antenna (5–7 dBi).

### GPS / GNSS Antenna

- Connect the **GPS patch antenna** to the U.FL/IPEX connector labeled `GNSS` on the board.
- Mount the antenna **horizontally** with a **clear view of the sky** for fastest satellite fix.
- Stick it to the top surface of the plastic enclosure, or route it outside the metal cabinet.

## Section 4 — Enclosure Mounting

1. Mount the Waveshare board inside an **IP65 plastic enclosure** using M3 standoffs or adhesive PCB standoffs.
2. Pre-drill cable glands for:
  - Mains power cable (to DIN-rail PSU → 5V to board)
  - PZEM power + serial cable (PZEM is usually mounted separately near the mains entry)
  - CT clamp cable
  - Antenna cable (if routing outside)
3. Mount the enclosure inside the electrical cabinet using double-sided mounting tape or DIN-rail clips.
4. **Do not use a metal enclosure for the ESP32 board** — it will block the 4G signal.

💡 The PZEM-004T module can be mounted separately inside the cabinet near the mains terminals, with only the 4-wire UART cable running to the ESP32 board enclosure.

# Section 5 — Power Supply

| Supply              | Specification                                      |
|---------------------|----------------------------------------------------|
| Mains Input         | 230V AC, 50Hz                                      |
| DIN-Rail PSU Output | 5V DC, ≥2A                                         |
| Board Supply        | Connect PSU 5V/GND to board's 5V header or USB-C   |
| PZEM Power          | Taken directly from mains terminals (self-powered) |

# Section 6 — LED Status Indicator

The onboard LED on GPIO 2 shows the current device state:

| LED Pattern                        | Meaning                            |
|------------------------------------|------------------------------------|
| Solid ON                           | Booting / Initializing             |
| Slow Blink (1 Hz)                  | Connecting to 4G cellular network  |
| Fast Blink (4 Hz)                  | Authenticating with Firebase cloud |
| Heartbeat (double-pulse, every 2s) | Normal operation — data uploading  |
| Rapid Flash                        | Error state — check serial monitor |

Next: [Firmware & Provisioning](#) →